

Sumit Pradeep Kolpe

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Education

University of Pennsylvania

Master of Science in Engineering in Data Science & AI (Fully Funded)

Expected Graduation: 05/28

Philadelphia, PA

- **Relevant Coursework:** Big Data Analytics, Databases & Information System, Statistics for Data Science.

Savitribai Phule Pune University

Bachelor of Engineering in Artificial Intelligence & Data Science

Graduation Date: 06/26

GPA: **3.9** / **4.0**

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Distributed Computing, Big Data Analytics, Statistics, Data Modeling & Visualization, Data Science, Machine Learning.

Experience

Avignon University, France — International AI/ML Research Intern

June 2025 – Aug 2025

- Achieved 95%+ F1-score in DoS detection by engineering a distributed intrusion-detection pipeline with PySpark and Hive, processing 100K+ network telemetry records and 15+ engineered traffic and performance features using Isolation Forest and XGBoost.
- Automated 50+ network simulation workflows using Python and OMNeT++, generating telemetry across varying traffic loads and attack intensities to quantify throughput, latency, packet loss, and detection response.
- Led a globally distributed team of 6 in developing a digital-twin-based network resilience framework, analyzing RIS configurations and AI-driven defense strategies; delivered the project with an A+ grade.

Infosys, India — Data Science & AI Intern

May 2024 – July 2024

- Reduced forecasting error by 20%+ by benchmarking AR, MA, ARMA, ARIMA, and SARIMA models using stationarity tests, ACF/PACF analysis, and walk-forward validation.
- Engineered 15+ temporal and business features from lag/rolling statistics, seasonality, pricing, promotions, and calendar effects to improve out-of-sample forecasting accuracy.
- Enhanced forecast explainability using SHAP and prediction intervals for transparent inventory planning; earned Infosys Best AI Tech Award at the Annual AI Summit for exceptional delivery and commitment.

Sponsorlytix, United Kingdom — Data Scientist Intern

Nov 2022 – Jan 2023

- Achieved 93%+ form-classification F1 with <100 ms/frame inference by engineering a real-time pose pipeline using OpenCV, MediaPipe landmarks, joint-angle features, and LSTM-based exercise recognition for posture correction and repetition counting.
- Improved workout-plan adherence by 18% by developing a personalized recommendation pipeline using biometric profiles, fitness goals, and longitudinal workout data to optimize exercise selection and progression.
- Increased workout engagement by 27% by engineering an event-driven pipeline converting workout activity into milestones, points, leaderboards, and redeemable rewards; won 1st place at an international hackathon and secured a published patent.

Relevant Project

Saheli: AI-Powered Women's Safety Platform | [Website](#)

Feb 2026

- Achieved 93%+ distress-detection precision by engineering an on-device keyword-spotting model using MFCC/log-Mel features and a lightweight CNN, deployed with TensorFlow Lite for low-latency, hands-free activation.
- Reduced alert-routing latency to <2 seconds by building an event-driven backend with FastAPI, PostgreSQL/PostGIS, Redis, and WebSockets for real-time location processing and proximity-based responder routing.
- Detected 8 high-risk zones and peak-risk periods by engineering 20+ spatiotemporal features with DBSCAN and XGBoost risk modeling, generating SHAP-based patrol recommendations; validated with 60 police departments and published an Indian patent.

Awards & Honors

- Earned National Aegis Graham Bell Award 2025, Ministry of Education, Government of India, for creating social impact through AI.
- Secured a fully funded merit-based Government of India Foreign Scholarship for graduate study at University of Pennsylvania.
- Ranked in the Top 10% among 57,000+ candidates nationwide in GATE-Data Science Exam 2025, qualifying on the first attempt.
- Selected among 30 finalist teams from 30 countries to represent India at the International Hackathon 2024 in Slovakia.
- Won 1st Prize among 452 teams at an international hackathon for developing a computer vision-based virtual gym trainer.

Leadership Experience

Institute Innovation Council (IIC), Government of India — Innovation Coordinator

Oct 2025 – May 2026

- Mentored 130+ student teams in translating technical ideas into socially responsible solutions across safety, healthcare, and sustainability by leading design-thinking and innovation initiatives.

NEURA: AI & Tech Club, India — President

Aug 2024 – Aug 2025

- Led a 110+ member AI & Data Science community and coordinated 3 major technical events, including the national-level "Dexterity," driving research, innovation, and interdisciplinary collaboration.

Skills

Python, SQL, PySpark, Apache Spark, Hive, Hadoop/HDFS, Pandas, NumPy, Statistical Modeling, Machine Learning, Scikit-learn, XGBoost, Time-Series Forecasting, Data Modeling, ETL, Data Pipelines, AWS, PostgreSQL, MongoDB, SHAP, TensorFlow, A/B Testing, FastAPI, Redis, PostGIS, Git, NLP, Power BI, Looker, Matplotlib, OpenCV, MediaPipe